International Journal of
Mental Health Nursing

International Journal of Mental Health Nursing (2019) ••, ••–•• doi: 10.1111/inm.12654

Review Article

The effectiveness of chemical restraint in managing acute agitation and aggression: A systematic review of randomized controlled trials

***Eimear Muir-Cochrane,*[1] *Candice Oster,*[1] *Adam Gerace,*[1,2] *Suzanne Dawson,*[1] *Raechel Damarell*[1] *and Karen Grimmer*[1]**

[1]*College of Nursing & Health Sciences, Flinders University, Adelaide, South Australia, and* [2]*School of Health, Medical and Applied Sciences, Central Queensland University, Rockhampton, Queensland, Australia*

ABSTRACT: *One approach to manage people with behaviours of concern including agitated or aggressive behaviours in health care settings is through the use of fast-acting medication, called chemical restraint. Such management often needs to be delivered in crisis situations to patients who are at risk of harm to themselves or others. This paper summarizes the available evidence on the effectiveness and safety of chemical restraint from 21 randomized controlled trials (RCTs) involving 3788 patients. The RCTs were of moderate to high quality and were conducted in pre-hospital, hospital emergency department, or ward settings. Drugs used in chemical restraint included olanzapine, haloperidol, droperidol, risperidol, flunitrazepam, midazolam, promethazine, ziprasidone, sodium valproate, or lorazepam. There was limited comparability between studies in drug choice, combination, dose, method of administration (oral, intramuscular, or intravenous drip), or timing of repeat administrations. There were 31 outcome measures, which were inconsistently reported. They included subjective measures of behaviours, direct measures of treatment effect (time to calm; time to sleep), indirect measures of agitation (staff or patient injuries, duration of agitative or aggressive episodes, subsequent violent episodes), and adverse events. The most common were time to calm and adverse events. There was little clarity about the superiority of any chemical method of managing behaviours of concern exhibited by patients in Emergency Departments or acute mental health settings. Not only is more targeted research essential, but best practice recommendations for such situations requires integrating expert input into the current evidence base.*

KEY WORDS: *aggression, agitation, chemical restraint, restraint, systematic review.*

Correspondence: Candice Oster, College of Nursing & Health Sciences, Flinders University, Adelaide, GPO Box 2100, Adelaide, SA 5001, Australia. Email: candice.oster@flinders.edu.au

Authorship statement: All authors listed meet the authorship criteria according to the latest guidelines of the International Committee of Medical Journal Editors. All authors are in agreement with the manuscript.

Disclosure statement: The authors declare no conflict of interest.

Eimear Muir-Cochrane, BSc Hons RN, Grad Dip Adult Ed., MNS, PhD, FACMHN, Credentialled Mental Health Nurse.
Candice Oster, BA(Hons), PhD.
Adam Gerace, BPsych(Hons), PhD.
Suzanne Dawson, MClinSc (Res), PGradDip, BAppSc (Occupational Therapy).
Raechel Damarell, BA, Grad Dip Info Stud, A.Mus.A..
Karen Grimmer, Cert Health Economics, BPhysiotherapy, PhD.

Accepted August 14 2019.

© 2019 Australian College of Mental Health Nurses Inc.

INTRODUCTION

Chemical restraint, also known as rapid tranquilisation, is the use of psychotropic medication to control severe agitation, or violent behaviours (Nadkarni *et al.* 2015). It is generally understood to be the *'assertive administration of emergency medication to adults with aggressive, agitated or violent behaviours with the purpose of quick calming or sedation, diminution of symptoms, and/or to decrease the likelihood of harm to self or others'* (Battagila *et al.* 2003, p.192). It may be administered in conjunction with physical restraint to induce a state of calm for people whose level of aggression poses

a threat to themselves or others, and which cannot be reduced by less invasive means (National Institute for Health and Care Excellence (NICE) guideline 2015).

Chemical restraint is most commonly used in pre-hospital settings, emergency departments (EDs), or acute psychiatric inpatient settings (Battaglia 2005). The practice is generally accepted as not being treatment, but rather to defuse acute symptoms in emergency situations and provide a safe space in which a diagnosis and treatment plan can then be determined (DHS Tasmania 2017). Consequently, chemical restraint can be lifesaving (Mott *et al.* 2005). However, it is acknowledged that medications given as chemical restraint may have more than the effect of controlling challenging behaviours and that these medications may also begin to treat symptoms of the underlying condition. Chemical restraint can furthermore be experienced as a violation of integrity and cause psychological discomfort (Haglund *et al.* 2003).

A search of international clinical practice guideline repositories found no evidence-based recommendations to inform the use by front-line clinicians of chemical restraint for management of severe agitation, aggression, or violent behaviours. Our recent systematic review of any literature on chemical restraint published since 1996 identified 33 potentially relevant systematic reviews (SRs) of primary studies (Muir-Cochrane *et al.* 2019). Only nine SRs included primary papers, which all specifically noted that medications had been administered without consent (Aguilera-Serrano *et al.* 2018; Goulet *et al.* 2017; Jarrett *et al.* 2008; Laiho *et al* 2013; Luciano *et al* 2014; Newton-Howes & Mullen 2011; Steinert *et al.* 2010; Tingleff *et al.* 2017; Weiland *et al.* 2017). None of these SRs reported on the effectiveness of medications. The remaining SRs all included primary papers in which people had provided consent, or not, for administration of medication for uncontrolled behaviours. Thus, these papers variably reported on emergency management, or treatment, of psychiatric disorders (Muir-Cochrane *et al.* 2019). There is therefore no current SR of literature dealing with the effectiveness of chemical restraint delivered to non-consenting adults to control violent behaviours or extreme agitation, and no guidance to front-line clinicians about current best practices.

AIMS

This review was informed by a previous investigation undertaken by the lead researcher with patients and carers about their understanding of, and concerns with, chemical restraint (Muir-Cochrane & Dawson 2016). The review aimed to describe current best evidence that could be adopted by front-line clinicians, such as nurses and doctors, when delivering and monitoring patients during chemical restraint for crisis management of non-consenting patients with behaviours of concern such as acute agitation, aggression, or violence.

Research question: *What is the effectiveness and safety of chemical restraint delivered by any route of administration (i.e. oral, Intramuscular (IM), or Intravenous (IV)), to manage non-consenting adults with behaviours of concern such as severe agitation, aggression, and violence?* For the purpose of this review, the Battagila *et al.* (2003) definition of chemical restraint (see above) was applied.

METHODS

The present review reports on randomized controlled trials (RCTs) that examined the effectiveness of medications used for chemical restraint in the management of acute agitation or aggression in acute hospital care, psychiatric or general hospital intensive care, EDs, or pre-hospital services. The RCTs were identified as a subset of the studies identified in a large systematic review on chemical restraint (Muir-Cochrane *et al.* 2019). Systematic reviews identified in the large review were used only as a source of RCTs for this study. The different study designs identified in the review were classified using the National Health and Medical Research Council hierarchy of evidence (Merlin *et al.* 2009) (Level II = RCTs; Level III-1 = prospective observational comparative studies; Level III-2 = cross-sectional observational studies; Level III-3 = retrospective studies; Level IV = opinion, case studies). The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Moher *et al.* 2015) was used to guide the methods for both the parent systematic review and subset reported here.

Search strategy and study selection

The parent data set (Muir-Cochrane *et al.* 2019) was collated to address a broad study question regarding available evidence on chemical restraint and involved a search of the databases PsycINFO, CINAHL, MEDLINE/PubMed, and Google Scholar. The search sought literature published since 1996 to ensure that changes over time in contemporary mental health practice could be examined. The search start date reflected findings in Jarrett *et al.*'s (2008) study of chemical restraint in acute care from 1980 to 2008 where only

© 2019 Australian College of Mental Health Nurses Inc.

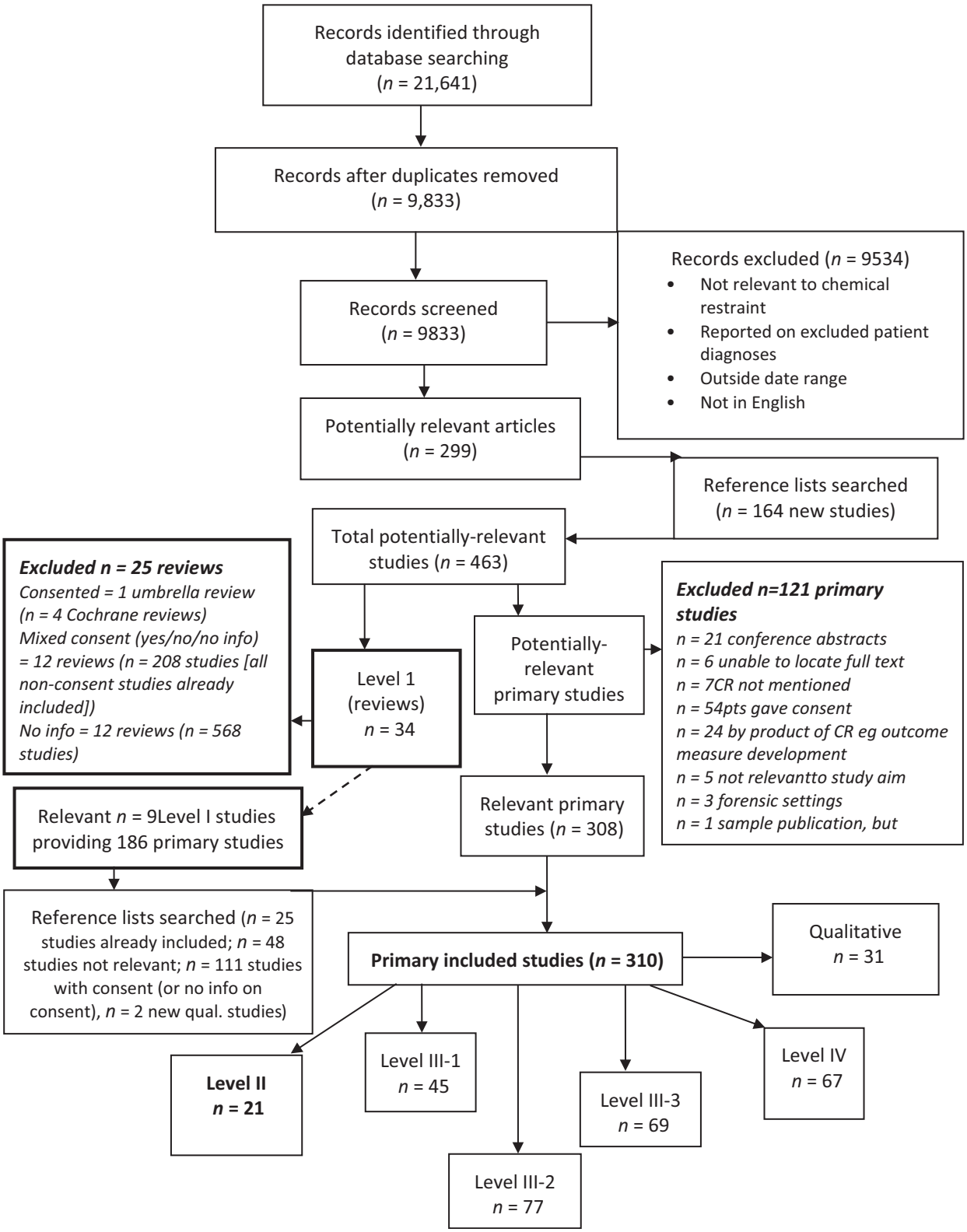

FIG. 1: *PRISMA flow diagram for the larger systematic review (Level II = RCTs; Level III-1 = prospective observational comparative studies; Level III-2 = cross-sectional observational studies; Level III-3 = retrospective studies; Level IV = opinion, case studies; RCT inclusions in bold).*

© 2019 Australian College of Mental Health Nurses Inc.

TABLE 1: *Summary of RCTs (organized by setting)*

Author (year) Country	Site	Analysed sample	Method	Outcome measure	Quality	Purpose	Results
Rosen *et al.* (1997)USA	Pre-hospital	46 combative patients with head trauma, medical conditions or psychiatric illness	IV	Agitation scale	13/13	Evaluate the effect of droperidol in managing combative patients (cf saline placebo) in pre-hospital situations	IV administration of 2–4 mg droperidol was a significantly better management option than saline placebo for combative behaviours
Isenberg and Jacobs (2015)USA	Pre-hospital	10 violent agitated patients	IM	Adverse physiological eventsRichmond Agitation & Sedation Scale	8/13	Evaluate efficacy of midazolam or haloperidol in sedating agitated pts	IM administration of midazolam and haloperidol is equally effective for sedating an agitated patient in the pre-hospital setting. Midazolam appears to have a faster onset of action, as evidenced by the shorter time required to achieve a RASS score of less than + 1 in the patients who received midazolam. Haloperidol offers an alternative option for the sedation of an agitated patient
Foster *et al.* (1997)USA	Psychiatric ED	37 acutely agitated patients	IM or oral	Brief Psychiatric Rating Scale (BPRS)Clinical Global Impressions (CGI)Length of Stay (LoS) ≥2/52	12/13	Compare neuroleptics with benzodiazepine for RT (administered orally or IM every 30 mins for 4 hours)	Oral or IM administrations of haloperidol or lorazepam had similar outcomes on producing rapid tranquilisation. However, lorazepam had less likelihood of producing side effects
Bieniek *et al.* (1998) USA	Psychiatric ED	20 acutely agitated patients	IM	CGIOvert Aggression Scale (OAS)VAS for agitation	13/13	Compare utility of lorazepam with combined haloperidol and lorazepam, to control acutely agitated behaviour	IM lorazepam combined with haloperidol was significantly better at calming aggression and agitation, than IM lorazepam alone
Alexander *et al.* (2004) India	Psychiatric ED	200 acutely agitated and violent patients	IM	Adverse physiological eventsAkathisia ScaleCGI-ImprovementCGI-SeveritySimpson–Argus ScaleTime to calm/sedation	12/13	Compare interventions for controlling agitation and violence in people with serious psychiatric disorders	IM haloperidol combined with promethazine acted more quickly, but no more effectively, than IM lorazepam in inducing calm or sleep for acutely agitated and violent patients
Baldacara *et al.* (2011) Brazil	Psychiatric ED	150 agitated and aggressive patients with psychotic or bipolar disorders	IM	OASOvert Agitation Severity Scale (OASS)Ramsey Sedation Scale (RSS)	13/13	Compare effectiveness of olanzapine, ziprasidone, or haloperidol (combined with midazolam or promethazine) in calming agitation and aggression	IM olanzapine, ziprasidone and haloperidol (alone, or combined with promethazine, or midazolam) were all effective in reducing agitation and aggression within 12 hours. However, haloperidol plus midazolam had the worst results in terms of unalleviated aggression, and the highest percentage of people with adverse events

(*Continued*)

© 2019 Australian College of Mental Health Nurses Inc.

TABLE 1: (Continued)

Author (year) Country	Site	Analysed sample	Method	Outcome measure	Quality	Purpose	Results
Huf *et al.* (2007) Brazil	Psychiatric ED	316 acutely agitated patients	IM	Adverse physiological eventsLoS ≥ 2/52Time to sedationUse of additional sedation	13/13	Compare speed of tranquilisation and safety of haloperidol administered alone or with promethazine	IM haloperidol combined with promethazine, was significantly more effective than IM haloperidol alone in producing RT within 20 mins, while minimizing side effects
Mantovani *et al.* (2013) Brazil	Psychiatric ED	100 patients with psychomotor agitation	IM	Agitation–Calmness Evaluating Scale (ACES)Adverse physiological eventsPositive & Negative Symptoms Scale (PANSS)	13/13	Test efficacy and safety of four IM low-dose antipsychotics for RT of psychomotor agitation	Low-dose IM haloperidol combined with midazolam, and low-dose IM olanzapine, were more effective than low-dose IM haloperidol & promethazine or low-dose IM ziprasidone, in reducing agitation without incurring adverse events for people with acute psychomotor agitation
Raveendran *et al.* (2007) India	Psychiatric ED	300 patients with agitation or aggression from mental illness	IM	% sedated in set time periodAdverse physiological eventsFurther coercive incidentsFurther violent episodesUse of additional sedation	13/13	Examine IM haloperidol & promethazine vs olanzapine for RT	IM haloperidol combined with promethazine and IM olanzapine were both effective than in producing RT within 15 mins, and neither incurred notable adverse events. More people treated with olanzapine needed additional sedation within four hours to maintain tranquilisation
TREC (2003) Brazil	Psychiatric ED	301 patients with aggression or agitation	IM	% sedated in set time periodAdverse physiological eventsFurther coercive incidentsFurther violent episodesLoS ≥ 2/52Use of additional sedation	13/13	Compare IM midazolam or IM haloperidol & promethazine for RT	Both IM haloperidol & lorazepam, and IM midazolam, were effective in producing rapid tranquilisation, however midazolam affected significantly more patients within 20 mins than haloperidol-lorazepam. Adverse events were rare for both treatment administrations
Asadollahi *et al.* (2015) Iran	ED	160 acutely agitated or violent patients	IV vs IM	Agitated Behavior Scale (ABS) ACESPANNS-Excited Component	10/13	Compare efficacy of valproate versus haloperidol in decreasing agitation level of affected pts	Both IM haloperidol and IV sodium valproate were effective in controlling behaviour within 30 min. IM haloperidol acted faster than IV sodium valproate, and however, IV sodium valproate was safer
Chan *et al.* (2013) Australia	ED	336 agitated patients	IV	% sedated in set time periodAdverse physiological eventsSedation scaleTime to calm/sedation	13/13	Determine efficacy and safety of droperidol or olanzapine as adjunct to midazolam	IV droperidol and IV olanzapine as adjuncts to IV midazolam were both more effective, and decreased the time to adequate sedation, compared with IV midazolam alone

(*Continued*)

© 2019 Australian College of Mental Health Nurses Inc.

TABLE 1: (Continued)

Author (year) Country	Site	Analysed sample	Method	Outcome measure	Quality	Purpose	Results
Knott *et al.* (2006) Australia	ED	153 acutely agitated patients	IV	% sedated in set time periodAdverse physiological eventsFurther violent episodesN. drugs adms to calm	13/13	Compare midazolam and droperidol for time to sedation and adverse events	IV midazolam and IM droperidol were equally effective in achieving sedation in 10 mins. More patients receiving midazolam were sedated within the first 5 min but more of these patients needed airways support, as well as further sedation at 60 mins (Knott et al., 2006). IV droperidol may be a safer option
Isbister *et al.* (2010) Australia	ED	91 patients with violent behaviours	IM	% sedated in set time periodAkathisia ScaleFurther violent episodesPatient injuriesSimpson–Argus ScaleStaff injuriesTime to calm/ sedationUse of additional sedation	13/13	Determine whether droperidol, midazolam or a combination is more effective for sedation	IM droperidol is a safer and quicker option to sedate violent and aggressive patients than IM midazolam. There was no additional benefit in using the combination IM administration of droperidol and midazolam
Martel *et al.* (2005)USA	ED	144 acutely agitated patients	IM	Altered Mental State (AMS) Oxygen saturationsPhysiology	13/13	Compare efficacy of sedation, need for rescue sedation, respiratory depression, complications of drugs to reduce acute undifferentiated agitation	IM midazolam and droperidol were faster acting and more effective than ziprasidone. Fewer patients given IM droperidol or IM ziprasidone required rescue medication to achieve adequate sedation, compared with midazolam. Ziprasidone delayed onset of adequate sedation the most. Respiratory distress was experienced by significantly more IM midazolam patients than the patients given the other drugs
Taylor *et al.* (2016)UK	ED	361 acutely agitated patients	IV	% sedated in set time periodAdverse physiological eventsTime to calm/sedation	13/13	Determine most efficacious of IV bolus for sedation, of midazolam and droperidol, OR droperidol OR olanzapine	IV administration of combined midazolam and droperidol produced significantly greater numbers of patients who were sedated within 10 min, compared with IV droperidol alone, or IV olanzapine alone. Considering the patients who were not adequately sedated in 10 min, the combined drug administration also required fewer additional doses to achieve sedation

(*Continued*)

© 2019 Australian College of Mental Health Nurses Inc.

TABLE 1: (Continued)

Author (year) Country	Site	Analysed sample	Method	Outcome measure	Quality	Purpose	Results
Yap *et al.* (2017) Australia	Psychiatric Intensive Care Unit (PICU)	92 agitated/ aggressive patients on meth-amphetamines	IV	% sedated in set time periodAdverse physiological eventsTime to calm/sedation	13/13	Examine the efficacy and safety of midazolam-droperidol versus droperidol, and midazolam-droperidol versus olanzapine for methamphetamine-related acute agitation.	Subset of Taylor et al., 2016:IV administration of combined midazolam and droperidol produced significantly greater numbers of patients who were sedated within 10 min, compared with IV droperidol alone, or IV olanzapine alone. Considering the patients who were not adequately sedated in 10 min, the combined drug administration also required fewer additional doses to achieve sedation
Calver *et al.* (2015) Australia	PICU	206 patients with aggressive behaviours	IM	% sedated in set time periodAdverse physiological eventsAkathisia ScalePatient injuriesSimpson–Argus ScaleStaff injuriesTime to calm/ sedationUse of additional sedation	13/13	Compare IM droperidol with IM haloperidol for effectiveness and safety for sedation for acute behavioural disturbance	IV administration of IM haloperidol (10 mg) or IM droperidol (10 mg) had similar effects in sedating patients with acute behavioural disorders, and however, the haloperidol arm was safer
Georgieva *et al.* (2013) Holland	Acute psychiatric ward	520 agitated and aggressive patients	Oral (if refused, IM)	PANNS	9/13	Evaluate whether seclusion or chemical restraints reduced aggression, where IM administration of chemical restraint was first choice of Rx	First choice involuntary use of IM medication, compared with first choice seclusion, did not reduce coercive events or use of seclusion to manage patients with aggressive or violent behaviours
Walther *et al.* (2014) Switzerland	Acute psychiatric ward	30 severely agitated patients with schizophrenia spectrum disorders	Oral	Time to calm/sedation	10/13	Investigate efficacy of oral haloperidol, risperidone, and olanzapine in reducing severe agitation over first 96 hours after admission	Effects of rapid oral tranquilisation (within two hours of first medication administration) and reduced psychotic agitation over a five-day period were similar for oral haloperidol (15 mg), oral olanzapine (20 mg) and oral risperidone (2–6 mg)
Dorevitch *et al.* (1999) Israel	IP	28 actively psychotic patients with acute agitation	IM	BPRSCGICGI-IFurther coercive incidentsOASSeclusionTime to calm/sedation	12/13	Examine the efficacy of IM flunirazepine compared with IM haloperidol to control aggression and/or agitation	IM flunitrazepam (1 mg) and IM haloperidol (5 mg) had similar effects on reducing aggression within 90 min of administration

ED, Emergency Department; IM, intramuscular injection; IP, inpatient; IV, intravenous administration; PICU, psychiatric intensive care unit; psych, psychiatric.

© 2019 Australian College of Mental Health Nurses Inc.

four papers were published pre-1996. Index and MeSH search terms were determined in consultation with a university liaison librarian and included chemical restraint, forced medication, coerced medication, intramuscular medication, rapid tranquilisation, restraint, mental health, psychiatric, emergency department, and variants/other terms identified in a preliminary review of the literature. Two independent reviewers screened the search results for eligibility.

To be included in the parent data set, articles were required to: (i) be peer-reviewed literature published in English reporting on the use of chemical restraint for non-consenting adults (with mental health diagnoses and/or substance abuse issues) exhibiting acute agitation, aggression, and/or violent behaviours in acute hospital care, psychiatric or general hospital intensive care, emergency departments, or pre-hospital services; (ii) articles of any research design were included in the parent data set, with only systematic reviews and randomized controlled trials addressing the effectiveness and safety of chemical restraint included in this analysis.

Studies were excluded if they were (i) published prior to 1996; (ii) short conference abstracts or non-peer-reviewed literature; or (iii) unavailable in English. Studies were also excluded if they reported on populations with specific contextual and/or health issues for which general chemical restraint recommendations may not be relevant (children or adolescents <18 years; patients with dementia, neurological diseases, or surgical and medical conditions; or patients in prison settings). Literature was also excluded if it was reported that subjects consented to administration of medications, or where medications were delivered specifically to treat agitation or aggression.

Reference lists of sourced articles were examined, and the primary literature contained within all included systematic reviews was scrutinized for RCTs not identified in the search. Studies were assessed for risk of bias by two independent reviewers using the Joanna Briggs Institute (JBI) RCT quality checklist (2014). The checklist contains thirteen items for assessing risk of bias in the studies, assessed by answering yes (Y), no (No) or unsure (U) against each item. The number of compliant items is summed, and the total reported. JBI do not provide guidance on an appropriate score to determine degrees of risk of bias, with varying approaches used by researchers (see e.g., Belay *et al.* 2019; Starbird *et al.* 2019). In this study, we determined a cut-off score for low risk of bias as papers with 10+ (>75%) compliant items. Non-compliant items were described.

Data extraction and synthesis

Data were extracted (as relevant) on study design, country of research, participant numbers and characteristics, site of research, year(s) in which research occurred, study purpose, definition of chemical restraint (if provided), reasons for chemical restraint, non-chemical constraint comparators, methods of administration, dosages and timing of intervention delivery, measures of outcome, summary information on baseline and post-test outcome measures, information on homogeneity at baseline between study arms, summary measures of effectiveness, and adverse events. The intention was to undertake a meta-analysis if two or more studies reported on similar patient populations, interventions, and outcomes. However, there was no consistency in use of outcome measures. Moreover, even if the same outcome measure was reported in two or more studies, it was rarely reported using the same summary statistics or measured over the same time frame. This precluded meta-analysis and effectiveness of interventions is therefore reported descriptively.

TABLE 2: *Timing of measurement of effect*

5–10 mins	20 mins	30 mins	1.5 hours	2 hours	3 hours	4 hours	12 hours
Rosen *et al.* (1997)	TREC (2003)	Asadollahi *et al.* (2015)	Mantovani *et al.* (2013)	Huf *et al.* (2007)	Bieniek *et al.* (1998)	Raveendran *et al.* (2007)	Baldacara *et al.* (2011)
Isenberg & Jacobs (2005)	Isbister *et al.* (2010)	Foster *et al.* (1997)		Martel *et al.* (2005)		Alexander *et al.* (2004)	
Knott *et al.* (2006)				Calver *et al.* (2015)			
Taylor *et al.* (2016)				Dorevitch *et al.* (1999)			
Yap *et al.* (2017)				Walther *et al.* (2014)			
Chan *et al.* (2013)							

Georgieva *et al.* (2013) did not provide relevant information.

© 2019 Australian College of Mental Health Nurses Inc.

RESULTS

The literature inclusion flow chart for the larger systematic review is reported in Figure 1, where the different types of study designs are indicated by the NHMRC hierarchy of evidence (Merlin *et al* 2009). There were 462 potentially relevant articles of all research hierarchies, including 33 systematic reviews (SRs) of primary research, and one umbrella review of four SRs. There were 19 SRs that synthesized evidence for different chemical restraint interventions, whose reference lists were searched for additional RCTs. In sum, 21 relevant RCTs were identified. Details on the excluded studies are from the authors by request.

The 21 included RCTs were conducted from 1997 to 2017 in nine countries: USA ($n = 5$), Australia ($n = 5$), Brazil ($n = 4$), India ($n = 2$), and one each from the United Kingdom, Iran, Israel, the Netherlands, and Switzerland. Study data were on average 2.8 years old (standard deviation (SD) 2.1) at time of publication (calculated as the difference from publication date to date of data collection, as cited in each paper). The studies reported, in total, on 51 treatment arms testing 3788 patients. One study (Yap *et al.* 2017) reported on a subset of 92 patients, which had already been included in the Taylor *et al.* (2016) data set of 361 patients; thus, the Yap *et al.* (2017) sample was not counted twice.

Study settings included pre-hospital ($n = 2$), psychiatric emergency departments ($n = 8$), general hospital emergency departments (EDs) ($n = 8$), psychiatric intensive care units (PICU) ($n = 2$), and acute psychiatric units ($n = 1$). The psychiatric conditions underlying agitated and aggressive behaviours were variably described. The lack of diagnoses in EDs was linked to staff capacity in that setting to assign a diagnosis. Control of symptoms was the primary aim in EDs in order to make decisions about whether to hospitalize people for diagnosis and treatment after aggression had subsided.

There was low risk of bias in 19 RCTs (scoring at least 75% of possible total 13 items). The remaining two studies had moderate to high risk of bias with scores of 9/13 (Georgieva *et al.* (2013), acute psychiatric ward, $n = 659$ patients) and 8/13 (Isenberg and Jacobs (2015), pre-hospital, $n = 10$ patients). The RCT total quality scores (out of 13) are reported in Table 1. The areas of least certainty identified during critical appraisal were how randomization to treatment groups had occurred, and whether (or how) patients and clinicians were blinded to treatment administration. Only five studies (Asadollahi *et al.* 2015; Calver *et al.* 2015; Isbister *et al.* 2010; Knott *et al.* 2006; Taylor *et al.* 2016) reported on the reference population as well as the number enrolled in the study; all others reported only on the number enrolled in the study. There were few drop outs because of study purpose and the timing of outcome assessment (outlined in Table 2); however, approximately 6% overall was lost from the studies because of errors in randomization, being found *post hoc* to be ineligible for the study, or because patients had been previously enrolled into another longer term trial and were thus on medications which made them ineligible for the chemical restraint trial. There was no evidence in any RCT that patients had been involved in study design or reporting. The included RCTs are summarized in Table 1.

Effectiveness of pharmacological interventions

All included RCTs shared a common purpose to test the effectiveness of pharmacological interventions for rapid, safe chemical restraint/rapid tranquilisation for people with uncontrolled agitation, aggression, or violence, who posed a danger to themselves and/or others. In line with the inclusion criteria, all RCTs overtly reported that patients were unable to provide informed consent at the time of drug administration and that chemical restraint was administered under duress to prevent injury to the patient or others. The rapidity of anticipated effect was highlighted by the timing of measurement of outcome.

Table 2 outlines the timing of measurement of effect over the included studies. Data were extracted from Walther *et al.* (2014) only for the rapid tranquilisation aspect of the study. In this study, once patients were well enough to provide informed consent, subsequent administration of medicines was for treatment, not behavioural control. Data were subsequently collected over 96 hours to assess the effectiveness of repeated sedation on symptoms of mental illness, agitation, aggression, and violence.

Interventions

In total, the 21 RCTs reported on 51 treatment arms.

The comparator arms were placebo or non-pharmacological interventions. Two studies reported on placebo arms (Rosen *et al.* (1997) reported using intravenous administration (IV) saline in a pre-hospital study, and Chan *et al.* (2013) reported on an IV saline

© 2019 Australian College of Mental Health Nurses Inc.

TABLE 3: *Study interventions and follow-up*

	Alexander	Asadollahi	Baldacara	Bieniek	Calver	Chan	Dorevitch	Foster	Georgieva
Method of administration	IM	IM	IM	IM	IM	IV	IM	IM or oral	Oral (if refused, IM)
Follow-up period	up to 4 hours	30 mins	up to 12 hours	3 hours	2 hours		2 hours	every 30 min for 4 hours	Not stated
Olanzapine			10 mg			5 mg			
Droperidol					10 mg	5 mg			
Risperidol									
Haloperidol		5 mg	5 mg		10 mg		5 mg	5 mg	
Haloperidol & Lorazepam				5 & 2 mg					
Haloperidol & Midazolam			5 mg & 15 mg						
Haloperidol & Promethazine	10 mg & 25–50 mg mix		5 mg & 50 mg						Oral: 10 mg & 100 mg OR if IM: 5 & 50 mg
Flunitrazepam							1 mg		
Midazolam						2.5 mg for < 50 kgs, 5 mg for> 50 kgs			
Droperidol & Midazolam									
Lorazepam	4 mg			2 mg				2 mg	Oral: 2.5–5 mg OR IM: 2.5–5 mg
Ziprasidone			20 mg						
Sodium Valproate		20 mg kg^{-1}							
Saline placebo						Not stated			
Seclusion									Isolation

Admins, administrations; IM, intramuscular; IV, intravenous administration.

drip in an ED study). Georgieva *et al.* (2013) used seclusion as a comparator.

Chemical restraint interventions: All but two studies (Georgieva *et al.* 2013; Rosen *et al.* 1997) compared two or more chemical restraint arms, largely testing equivalence hypotheses. Thirteen studies tested chemical restraint administered via intramuscular injection (IM), two studies tested chemical restraint administered in two ways (orally or IM), one study tested oral chemical restraint only, and five studies tested intravenous administration (IV). Chemical restraint (tested alone, or in combination) comprised olanzapine, haloperidol, droperidol, risperidol, flunitrazepam, midazolam, ziprasidone, sodium valproate, and lorazepam. The most commonly tested individual chemical restraints were droperidol and haloperidol (eight study arms each), followed by olanzapine, or haloperidol combined with promethazine (seven study arms each).

© 2019 Australian College of Mental Health Nurses Inc.

Huf	Isbister	Isenberg	Knott	Mantovani	Martel	Raveendran	Rosen	Taylor	TREC	Walther	Yap
IM	IM	IM	IV	IM	IM	IM	IV	IV	IM	Oral	IV
120 min	Time to calm	every 5 min	5 & 10 mis	90 min	up to 120 min	up to 240 min	10 min	10 min	20 min	2 hours	10 min
				10 mg		10 mg		10 mg		20 mg daily	10 mg
	10 mg		5 mg every 5 mins		5 mg		5 mg	10 mg			10 mg
										2–6 mg daily	
5 mg		5 mg if < 65 yrs, 2.5 mg if 65 + years								15 mg daily	
				2.5 mg & 7.5 mg							
5 mg & up to 50 mg				2.5 mg & 25 mg		10 mg & 25 or 50 mg			5–10 mg & 50 mg		
	10 mg	5 mg if < 65 years, 2.5 mg if 65 + years	5 mg every 5 mins		5 mg				15 mg		
	50 & 5 mg							5 & 5 mg			5 & 5 mg
				10 mg	20 mg						
							5 mg				

There was limited comparability between studies in chemical restraint choice, dose, method of administration, or timing of repeat administrations. Table 3 reports on the study interventions (chemical restraint, dosage, method of administration), and the follow-up period for measurement of effect.

Outcome measures: Outcomes varied and included subjective clinician measures of aggression and agitation, aggressive incidents or injuries, objective measures of sedation, adverse events, physiological measures, and service delivery measures. The need for additional sedation to achieve the chosen outcome measure was collected and reported in 20/21 RCTs. Adverse events (AE) were also reported in 20/21 studies, with 19 studies describing the AEs collected. There was no standard way of measuring or reporting AEs (see Table 4). The frequency of AE differed between studies (ranging from 0% (Bieniek *et al.* 1998;

© 2019 Australian College of Mental Health Nurses Inc.

TABLE 4: *Reporting need for additional sedation, description of adverse events (AE), and percentage of total subjects reporting AE*

Setting	Author, Year	Additional sedation	Adverse events (AE) description	% AE in total sample
Pre-hospital	Rosen 1997	Y	Not described	0
Pre-hospital	Isenberg 2015	Y	Need for intubationTime between the start of the Q wave and the end of the T wave in the heart's electrical cycle (QT interval) greater than 500 millisecondsCardiac arrhythmiasTime until the patient was awake for discharge	0
Psych ED	Foster 1997	Y	Blood pressure changesExtreme sedation	13.5
Psych ED	Bieniek 1998	Y	Adverse events not specified	0
Psych ED	TREC Group 2003	Y	Severe adverse events not specified	0.8
Psych ED	Alexander 2004	Y	Simpson–Angus extrapyramidal side effects rating scaleBarnes Akathisia ScaleOther adverse effects (esp dystonia)	1.0
Psych ED	Huf 2007	Y	As defined by the front-line clinicians	3.8
Psych ED	Raveendran 2007	Y	Simpson–Angus extrapyramidal side effects rating scale, Barnes Akathisia Scale, dystonia	1.3
Psych ED	Baldacara 2011	Y	Akathisia, dystonic reactions, amnesia, respiratory depression, paradoxical reactions and confusion and neuroleptic malignant syndrome	8.7
Psych ED	Mantovani 2013	Y	Ugvalg Klinisk Undersgelser Side Effects Scale	31.0
ED	Martel 2005	Y	Hypotension, seizures, dystonia, akathisia, dysrhythmia (QTprolongation and torsades de pointes)	14.6
ED	Knott 2006	Y	Active airway management (eg, jaw thrust, oral or nasal airway), assistance with ventilation (eg, bag and mask), oxygen required for documented desaturation below 90%, systolic blood pressure less than 90 mm Hg, documented arrhythmia, dystonic reaction, seizure, vomiting, or aspiration of stomach contents.	13.7
ED	Isbister 2010	Y	Respiratory depression requiring intubation, arrhythmias including torsades des pointes, extrapyramidal adverse effects requiring administration of benztropine, anaphylaxis, or any other major serious unexpected effect	13.2
ED	Chan 2013	Y	Airway obstruction, Oxygen desaturation, Hypotension§ Arrhythmia Decreased Glasgow Coma Scale	11.6
ED	Asodollahi 2015	Y	Cardiac problems, intense sedative effect, extrapyramidal symptoms, (EPS) (dystonia and akathisia), and anticholinergic side effects deep sedation	38.1
ED	Taylor 2016	Y	Airway compromise, oxygen desaturation, hypotension, extrapyramidal events	19.5
ED	Yap 2017†	Y	Airway compromise, oxygen desaturation, hypotension, extrapyramidal events	16.3
PICU	Dorevitch 1999	Y	Acute extrapyramidal events	0
PICU	Georgieva *et al.* (2013)	Not reported	Not reported	
PICU	Walther 2014	Y	Motor adverse events measured by BARS, SAS and AIMS	Not reported
PICU	Calver 2015	Y	Respiratory rate less than 12 breaths min^{-1}; systolic blood pressure less than 90 mmHg; heart rate less than 60 beats min^{-1}; oxygen saturation less than 90%; presence of extrapyramidal side effects	3.1

†Subset of Taylor *et al.* (2016).

Dorevitch *et al.* 1999; Isenberg & Jacobs 2015; Rosen *et al.* 1997) to 38.1% (Asodollahi *et al.* 2015)).

While the general purpose of all RCTs was to manage uncontrolled agitation, aggression, and/or violent behaviours in crisis situations (see Table 1), end-points were variably expressed as calm, sedation, or a specified amount of change on a subjective aggression measure. Calm and sedation were variably assessed as different states of arousability, or as a category in a subjective sedation outcome measure. Considering outcome measures associated with sedation, the most common measure (in 11 studies) was 'time to calm' although the state of 'calm' was variably described, and the next most common measure was the percentage of people sedated within a time period (usually 5 or 10 min; $n = 8$ studies).

© 2019 Australian College of Mental Health Nurses Inc.

TABLE 5: *Synthesis of results*

Setting	Author (year)	Measure of control of agitation, aggression or violent behaviours		Minimize adverse events
Pre-hospital	Rosen *et al.* (1997)	Effect	**IV 2–4 mg DPD** ↑ saline	
	Isenberg and Jacobs (2015)	Time to act	**IM MDZ 5 mg** ↑ **IM HPL 5 mg**	**IM MDZ** ↔ **IM HPL**
	Isenberg and Jacobs (2015)	Time to normalize	**IM HPL** ↑ **IM MDZ**	
Psychiatric ED	Foster *et al.* (1997)	Effect	**Oral or IM 5 mg HPL** ↔ **IM or oral 2 mg LZM**	**LZP** ↑ HPL
	Bieniek *et al.* (1998)	Effect	**IM 5 mg HPL & 2 mg LZP** ↑ IM 2 mg LZP	**IM HPL & LZP** ↔ **IM LZP**
	TREC (2003)	Speed	**IM 15 mg MDZ** ↑ IM 5–10 mg HPL & 50 mg LZP	**IM HPL & LZP** ↔ **IM MDZ**
	TREC (2003)	Effect	**IM 5–10 mg HPL & 50 mg LZP** ↔ **IM 15 mg MDZ**	
	Alexander *et al.* (2004)	Effect	**IM 10 mg HPL & 25–50 mg PMZ** ↔ **IM 4 mg LZP**	
	Huf *et al.* (2007)	Speed	**IM 5 mg HPL & (up to) 50 mg PMZ** ↑ IM 5 mg HPL	**IM HPL & PMZ** ↑ IM HPL
	Huf *et al.* (2007)	Effect	**IM 5 mg HPL & (up to) 50 mg PMZ** ↔ **IM 5 mg HPL**	
	Raveendran *et al.* (2007)	Effect	**IM 10 mg HPL & 25–50 mg PMZ** ↔ **IM 10 mg OLZ**	**IM HPL & PMZ** ↑ IM OLZ
	Baldacara *et al.* (2011)	Effect	**IM 5 mg HPL** ↔ **IM 5 mg HPL & 50 mg PMZ** ↔ **IM 5 mg HPL & 15 mg MDZ** ↔ **IM 10 mg OLZ** ↔ **IM 20 mg ZSD**	**(IM HPL** ↔ **IM HPL & PMZ** ↔ **IM OLZ** ↔ **IM ZSD)** ↑ IM HPL & MDZ
	Mantovani *et al.* (2013)	Effect	**(IM 2.5 mg HPL & 7.5 mg MDZ** ↔ **IM 10 mg OLZ)**↑ (IM 2.5 mg HPL & 25 mg PMZ ↔ IM 10 mg ZSD)	(**IM HPL & MDZ** ↔ **IM OLZ**) ↑ (IM HPL & PMZ ↔ IM ZSD)
	Richards et al. (1998)	Time to act	**IV 2.5–5 mg DPD** ↑ IV 2.5–5 mg HPL	
ED	Chan *et al.* (2013)	Time to act	**(IV DRP 5 mg & MDZ 2.5 mg** ↔ **IV OLZ 5 mg & MDZ 2.5 mg)**↑IV MDZ 2.5 mg& saline	IM DRD 5 mg & MDZ 2.5 mg ↔ IM OLZ 5 mg & MDZ 2.5 mg ↔ MDZ 2.5 mg
	Martel *et al.* (2005)	Time to act	**(IM 5 mg MDZ** ↔ **IM 5 mg DPD)** ↑IM 20 mg ZSD	**(IM DPD** ↔ **IM ZSD)** ↑IM MDZ
	Martel *et al.* (2005)	Effect	**(IM 5 mg MDZ** ↔ **IM 5 mg DPD)** ↑IM 20 mg ZSD	
	Knott *et al.* (2006)	Time to act	**IV 5 mg per 5 mins MDZ** ↔ **IV 5 mg per 5 mins DPD**	**IV DPD** ↑IV MDZ
	Knott *et al.* (2006)	Effect	**IV 5 mg per 5 mins MDZ** ↔ **IV 5 mg per 5 mins DPD**	
	Isbister *et al.* (2010)	Time to act	**IM 10 mg DPD** ↑IM 10 mg MDZ alone or IM 5 mg DPD & 5 mg MDZ	**IM DPD** ↑IM MDZ alone or IM PD & MDZ
	Isbister *et al.* (2010)	Effect	**IM 10 mg DPD** ↑IM 10 mg MDZ alone or IM 5 mg DPD & 5 mg MDZ	
	Asadollahi *et al.* (2015)	Time to act	**IM 5 mg HPD** ↑IV 20 mg kg^{-1} NaV	**IV NaV** ↑ IM HPD
	Asadollahi *et al.* (2015)	Effect	**IM 5 mg HPD** ↔ IV 20 mg kg^{-1} NaV	
	Taylor *et al.* (2016(	Time to act	**IV 5 mg MDZ & 5 mg DPD** ↑ (IV 10 mg DPD or IV 10 mg OLZ)	**IV MDZ & DPD** ↑ (IV DPD or IV OLZ)
	Taylor *et al.* (2016)	Effect	**IV 5 mg MDZ & 5 mg DPD** ↑ (IV 10 mg DPD or IV 10 mg OLZ)	
Psychiatric Intensive Care Unit (PICU)	Yap *et al.* (2017)	Time to act	**IV 5 mg MDZ & 5 mg DPD** ↑ (IV 10 mg DPD or IV 10 mg OLZ)	**IV MDZ & DPD** ↑ (IV DPD or IV OLZ)

(*Continued*)

© 2019 Australian College of Mental Health Nurses Inc.

TABLE 5: (Continued)

Setting	Author (year)	Measure of control of agitation, aggression or violent behaviours		Minimize adverse events
	Yap *et al.* (2017)	Effect	**IV 5 mg MDZ & 5 mg DPD** ↑ (IV 10 mg DPD or IV 10 mg OLZ)	
	Calver *et al.* (2015)	Time to act	**IM 10 mg HPL ↔ IM 10 mg DPD**	**IM HPL** ↑ IM DPD
	Calver *et al.* (2015)	Effect	**IM 10 mg HPL ↔ IM 10 mg DPD**	
Acute psychiatric ward	Walther *et al.* (2014)	Time to act	**oral 15 mg HPL ↔ oral 20 mg OLZ ↔ oral 2 –6 mg RPD**	
	Walther *et al.* (2014)	Effect	**oral 15 mg HPL ↔ oral 20 mg OLZ ↔ oral 2 –6 mg RPD**	
	Dorevitch *et al.* (1999)	Time to act	**IM 1 mg FZM ↔ IM 5 mg HPL**	
	Dorevitch *et al.* (1999)	Effect	**IM 1 mg FZM ↔ IM 5 mg HPL**	

The bolded drugs highlight significant within arm differences, and all treatments within brackets produced similar outcomes. The symbol ↑ indicates a significantly better treatment outcome (effect or speed of effect), and the symbol ↔ indicates equivalence; DPD, droperidol; FZM, flunitrazepam; HPL, The drug names are shortened haloperidol; LZP, lorazepam; MDZ, midazolam; NaV, sodium valproate; OLZ, olanzapine; PMZ, promethazine; RPD, risperidone; ZSD, ziprasidone.

Synthesis of results

The overall results of the intervention arms are summarized in Table 5, in terms of comparative effectiveness between intervention arms (irrespective of the measure of outcome) and (if relevant) speed of effectiveness. This table also reports on AEs. Blank cells indicate non-reporting of AEs in separate study arms. The Georgieva *et al.* (2013) study was not included in Table 5 because it did not measure effect or speed of treatment or AEs.

Considering the many methods of chemical restraint, haloperidol was the most common choice (alone or combined with other drugs). Chemical restraint of any type can be safely administered by IV with a significant calming effect observed within 10 mins of drug administration. Oral and IV drug administration of chemical restraint appear to be as effective as IM drug delivery. Only two studies compared chemical restraint with non-pharmacological interventions (saline (Rosen *et al.* 1997); seclusion Georgieva *et al.* (2013)). Because of the equivocality comparisons in the remaining studies, it was not possible to determine whether one drug was more effective than any other. Midazolam appeared to act more quickly than any other drug to control symptoms, but its effect was shorter lasting (e.g. 30 min). Midazolam also appears to incur more adverse events than other drugs (such as heavy sedation, reduced arousability, or respiratory depression). There was inconclusive benefit of combining drugs to enhance speed or size of effect on agitation, aggression, or violence, although the combination of midazolam and droperidol was reported as more effective in terms of rapid sedation for patients with methamphetamine-related acute agitation, than droperidol or olanzapine alone (Yap *et al.* 2017).

DISCUSSION

This paper provides the first systematic review of randomized controlled trials (RCTs) reporting on any form of chemical restraint, for non-consenting people with uncontrolled aggression, violence, or agitation from psychiatric disorders and/or substance abuse issues. It provides the first comprehensive guidance that we are aware of for front-line clinicians regarding choice of drug for rapid tranquilisation. The need to maintain safety for all concerned when caring for people with behaviours of concern, such as aggression or agitation that may lead to harm, is of foremost concern for health professionals. This systematic review of the effectiveness and safety of chemical restraint details findings from 21 RCTs designed specifically to test the efficacy of chemical restraint in acute psychiatric circumstances. While the trials demonstrate that the drugs tested (alone or in combination) generally rapidly reduce the symptoms of concern, this review of 22 years of research highlights the ongoing lack of consensus regarding best practice.

Lack of clarity in outcome measures regarding the endpoint of treatment made it difficult to combine findings from the RCTs. For instance, 11 papers reported on

© 2019 Australian College of Mental Health Nurses Inc.

objective measures of success such as time to calm or sedation, and eight papers reported on the percentage of patients achieving calm or sedation within a specific time period. However, calm is a different state to sedation and occurs prior to sedation, and Garriga *et al.* (2016) recommend that chemical restraint ideally should calm and not over sedate. The majority of papers (15/21) reflected pre-hospital and emergency department settings, and not surprisingly, these papers focused on time to control symptoms or the percentage of patients whose aggression, agitation, or violence was controlled within specific time periods (5–10 min). Given chemical restraint is administered '*with the purpose of quick calming or sedation, diminution of symptoms, and/or to decrease the likelihood of harm to self or others*' (Battagila *et al.* 2003, p.192), these outcomes appeared to be more appropriate and useful for front-line clinicians than was available from subjective outcome measures.

Despite the range of outcome measures used, the range of drugs, dosages, and methods of administration, control over aggression, agitation, or violence can be gained quickly once drugs are administered. The heterogeneity of interventions and outcome measures precluded meta-analysis; however, it appeared that haloperidol or droperidol (alone or in combination with other drugs) were common, safe, and effective choices compared with other options.

From the included studies, it was not possible to establish the most effective or safe drug (or dose of drug), or administration route, for chemical restraint. All but two of the RCTs reported on equivocality trials of different types, combinations, and/or doses of drugs, and thus could only indicate whether one intervention improved the chosen outcome measures compared to others. Of the two studies that compared chemical restraint with non-pharmacological interventions, chemical restraint appeared to be more effective in terms of speed and reduction of symptoms. The common finding of adverse events for most forms of chemical restraint highlights the importance that chemical restraint be used as a measure of last resort in managing behaviours of concern (Garriga *et al.* 2016). This supports the need for close monitoring of chemically restrained patients. The medical monitoring of patients in EDs is evident in RCTs compared with recent evidence reporting little documentation of side effect monitoring in adult psychiatric inpatient units (Hu *et al.* 2019).

For a clearer picture to be developed of best practice chemical restraint administration, it seems that agreement should be sought on standard protocols for drug choice, dose, and method of administration for crisis management of challenging behaviours, as well as a standard battery of outcome measures. Further, the individual experience of any form of restraint is well recognized as traumatizing and anti-therapeutic (Brophy *et al.* 2016). Research undertaken by Georgieva *et al.* (2012) found that patient preferences in regard to different forms of restraint were defined by previous experiences; those who had not been previously restrained and those that had experienced seclusion and chemical restraint together preferred to be medicated in an emergency. Patient preference regarding various forms of restraint is thus an important consideration to reduce the trauma experienced and preferences could be indicated through safety plans and advanced directives. However, this may be hard to operationalize in emergency situations where patients may be at risk of harm to self or others.

As with all systematic reviews of the literature, while we attempted to apply as comprehensive a search strategy as possible, there remains the potential that relevant RCTs were not identified. Moreover, the exclusion of unpublished literature potentially introduced bias by excluding potentially relevant studies. However, we believe that at this point in time we have produced the most comprehensive compilation of literature in this area, sufficient to provide guidance to front-line clinicians when making choices about chemical restraint options.

CONCLUSION

Our review demonstrates sustained interest over 22 years in the best ways to crisis manage psychiatric patients with challenging behaviours presenting to healthcare settings, when non-consenting chemical restraint is indicated. While there is a sizeable, good quality body of RCT evidence regarding chemical restraint practices from around the world, the interventions, outcome measures, and findings are heterogenous and preclude more than simple description. On the current evidence base from RCTs, front-line clinicians could be advised to use haloperidol (alone or in combination with lorazepam or midazolam) delivered via IV, oral, or IM methods, to safely, speedily, and effectively control agitated, aggressive, and violent behaviours.

RELEVANCE FOR CLINICAL PRACTICE

Use of chemical restraint as a last resort intervention can be required to maintain the safety of patients and staff. Side effects monitoring by nurses and other clinical staff and physical observations and documentation

© 2019 Australian College of Mental Health Nurses Inc.

are vital to ensure the health status of patients who have been chemically restrained. Clinicians need to be up to date about the latest evidence regarding medications and have specific protocols for the use of chemical restraint.

ACKNOWLEDGEMENTS

Funding was received from a Flinders University School of Nursing and Midwifery Start-Up Grant.

REFERENCES

Aguilera-Serrano, C., Guzman-Parra, J., Garcia-Sanchez, J. A., Moreno-Kustner, B. & Mayoral-Cleries, F. (2018). Variables associated with the subjective experience of coercive measures in psychiatric inpatients: A systematic review. *Canadian Journal of Psychiatry*, *63*, 129–144.

Alexander, J., Tharyan, P., Adams, C., John, T., Mol, C. & Philip, J. (2004). Rapid tranquillisation of violent or agitated patients in a psychiatric emergency setting. *British Journal of Psychiatry*, *185*, 63–69.

Asadollahi, S., Heidari, K., Hatamabadi, H., *et al.* (2015). Efficacy and safety of valproic acid versus haloperidol in patients with acute agitation: Results of a randomized, double-blind, parallel-group trial. *International Clinical Psychopharmacology*, *30*, 142–150.

Baldacara, L., Snches, M., Cordeiro, D. C. & Jackowski, A. P. (2011). Rapid tranquilization for agitated patients in emergency psychiatric rooms: A randomized trial of olanzapine, ziprasidone, haloperidol plus promethazine, haloperidol plus midazolam and haloperidol alone. *Revista Brasileira de Psiquiatria*, *33*, 30–39.

Battaglia, J. (2005). Pharmacological management of acute agitation. *Drugs*, *65*, 1207–1222.

Battaglia, J., Lindborg, S. R., Alaka, K., Meehan, K. & Wright, P. (2003). Calming versus sedative effects of intramuscular olanzapine in agitated patients. *American Journal of Emergency Medicine*, *21*, 192–198.

Belay, G. M., Endalamaw, A. & Ayele, A. D. (2019). Late presentation of HIV positive adults and its predictors to HIV/AIDS care in Ethiopia: A systematic review and meta-analysis. *BMC Infectious Diseases*, *19*, 534.

Bieniek, S. A., Ownby, R. L., Penalver, A. & Dominguez, R. A. (1998). A double-blind study of Lorazepam versus the combination of Haloperidol and Lorazepam in managing agitation. *Pharmacotherapy*, *18*, 57–62.

Brophy, L. M., Roper, C. E., Hamilton, B. E., Tellez, J. J. & McSherry, B. M. (2016). Consumers and carer perspectives on poor practice and the use of seclusion and restraint in mental health settings: Results from Australian focus groups. *International Journal of Mental Health Systems*, *10*, 6.

Calver, L., Drinkwater, V., Gupta, R., Page, C. B. & Isbister, G. K. (2015). Droperidol v. Haloperidol for sedation of aggressive behaviour in acute mental health: Randomised controlled trial. *The British Journal of Psychiatry*, *206*, 223–228.

Chan, E. W., Taylor, D. M., Knott, J. C., Phillips, G. A., Castle, D. J. & Kong, D. C. M. (2013). Intravenous Droperidol or Olanzapine as an adjunct to Midazolam for the acute agitated patient: A multicentre, randomized, double-blind, placebo-controlled clinical trial. *Annals of Emergency Medicine*, *61*, 72–81.

Dorevitch, A., Katz, N., Zemishlany, Z., Aizenberg, D. & Weizman, A. (1999). Intramuscular flunitrazepam versus intramuscular haloperidol in the emergency treatment of aggressive psychotic behavior. *American Journal of Psychiatry*, *156*, 142–144.

Foster, S., Kessel, J., Berman, M. E. & Simpson, G. M. (1997). Efficacy of lorazepam and haloperidol for rapid tranquilization in a psychiatric emergency room setting. *International Clinical Psychopharmacology*, *12*, 175–179.

Garriga, M., Pacchiarotti, I., Kasper, S., *et al.* (2016). Assessment and management of agitation in psychiatry: Expert consensus. *The World Journal of Biological Psychiatry*, *17*, 86–128.

Georgieva, I., Mulder, C. L. & Wierdsma, A. (2012). Patients' preference and experiences of forced medication and seclusion. *Psychiatric Quarterly*, *83*, 1–13.

Georgieva, I., Mulder, C. L. & Noorthoorn, E. (2013). Reducing seclusion through involuntary medication: A randomized clinical trial. *Psychiatry Research*, *205*, 48–53.

Goulet, M.-H., Larue, C. & Dumais, A. (2017). Evaluation of seclusion and restraint reduction programs in mental health: A systematic review. *Aggression and Violent Behavior*, *34*, 139–146.

Haglund, K., Von Knorring, L. & Von Essen, L. (2003). Forced medication in psychiatric care: Patient experiences and nurse perceptions. *Journal of Psychiatric and Mental Health Nursing*, *10*, 65–72.

Hu, F., Muir-Cochrane, E., Oster, C. & Gerace, A. (2019). An examination of the incidence and nature of chemical restraint on adult acute psychiatric inpatient units in Adelaide, South Australia. *International Journal of Mental Health Nursing*, https://doi.org/10.1111/inm.12591.

Huf, G., Coutinho, E. S. F. & Adams, C. E. (2007). Rapid tranquillisation in psychiatric emergency settings in Brazil: Pragmatic randomised controlled trial of intramuscular haloperidol versus intramuscular haloperidol plus promethazine. *BMJ*, *335*, 1–7.

National Institute for Health and Care Excellence (NICE) (2015). Violence and Aggression: Short-Term Management in Mental Health, Health and Community Settings. Updated edition. https://www.ncbi.nlm.nih.gov/books/NBK356335/#ch2.s14. Accessed 2 August 2019.

Isbister, G. K., Calver, L., Page, C. B., Stokes, B., Bryant, J. L. & Downes, M. A. (2010). Randomized controlled trial of intramuscular droperidol versus midazolam for violence and acute behavioral disturbance: The DORM study. *Annals of Emergency Medicine*, *56*, 392–401.

Isenberg, D. L. & Jacobs, D. (2015). Prehospital Agitation and Sedation Trial (PhAST): A randomized control trial of intramuscular haloperidol versus intramuscular midazolam

© 2019 Australian College of Mental Health Nurses Inc.

for the sedation of the agitated or violent patient in the prehospital setting. *Prehospital Disaster Medicine*, *30*, 491–495.

Jarrett, M., Bowers, L. & Simpson, A. (2008). Coerced medication in psychiatric inpatient care: Literature review. *Journal of Advanced Nursing*, *64*, 538–548.

Knott, J. C., McD Taylor, D. & Castle, D. (2006). Randomized clinical trial comparing intravenous midazolam and droperidol for sedation of the acutely agitated patient in the emergency department. *Annals of Emergency Medicine*, *47*, 61–67.

Laiho, T., Kattainen, E., Astedt-Kurki, P., Putkonen, H., Lindberg, N. & Kylma, J. (2013). Clinical decision making involved in secluding and restraining an adult psychiatric patient: An integrative literature review. *Journal of Psychiatric and Mental Health*, *20*, 830–839.

Luciano, M., Sampogna, G., del Vecchio, V., *et al.* (2014). Use of coercive measures in mental health practice and its impact on outcome: A critical review. *Expert Review of Neurotherapeutics*, *14* (2), 131–141.

Mantovani, C., Labate, C. M., Sponholz, A., *et al.* (2013). Are low doses of antipsychotics effective in the management of psychomotor agitation? A randomized, rate-blind trial of 4 intramuscular interventions. *Journal of Clinical Psychopharmacology*, *33*, 306–312.

Martel, M., Sterzinger, A., Miner, J. R., Clinton, J. & Biros, M. (2005). Management of acute undifferentiated agitation in the emergency department: A randomized double-blind trial of droperidol, siprasidone, and midazolam. *Academic Emergency Medicine*, *12*, 1167–1172.

Merlin, T., Weston, A. & Tooher, R. (2009). Extending an evidence hierarchy to include topics other than treatment: Revising the Australian 'levels of evidence'. *BMC Medical Research Methodology*, *9*, 34.

Moher, D., Shamseet, L., Clarke, M., *et al.* (2015). Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews*, *4*, 1.

Mott, S., Poole, J. & Kenrick, M. (2005). Physical and chemical restraints in acute care: Their potential impact on the rehabilitation of older people. *International Journal of Nursing Practice*, *11*, 95–101.

Muir-Cochrane, E. C. & Dawson, S. K. (2016). Examination of the definition and use of chemical restraint in acute psychiatric settings. The MHS Conference 2016. Auckland, New Zealand. Aug 2016.

Muir-Cochrane, E., Oster, C. & Grimmer, K. (2019). International research into 22 years of chemical restraint: An evidence overview. *Journal of Evaluation in Clinical Practice*. https://doi.org/10.1111/jep.13232

Nadkarni, P., Jayaram, M., Nadkarni, S., Rattehalli, R. & Adams, C. E. (2015). Rapid tranquillisation: A global perspective. *BJPsych International*, *12*, 100–102.

Newton-Howes, G. & Mullen, R. (2011). Coercion in psychiatric care: Systematic review of correlates and themes. *Psychiatric Services*, *62*, 465–470.

Raveendran, N. S., Tharyan, P., Alexander, J. & Adams, C. E. (2007). Rapid tranquillisation in psychiatric emergency settings in India: Pragmatic randomised controlled trial of intramuscular olanzapine versus intramuscular haloperidol plus promethazine. *British Medical Journal*, *335*, 869.

Richards, J. R., Derlet, R. & Duncan, D. R. (1998). Chemical restraint for the agitated patient in the emergency department: Lorazepam versus droperidol. *The Journal of Emergency Medicine*, *16*, 567–573.

Rosen, C. L., Ratliff, A. F., Wolfe, R. E., Branney, S. W., Roe, E. J. & Pons, P. T. (1997). The efficacy of intravenous droperidol in the prehospital setting. *The Journal of Emergency Medicine*, *15*, 13–17.

Starbird, L. E., DiMaina, C., Sun, C.-A. & Han, H.-R. (2019). A systematic review of interventions to minimize transportation barriers among people with chronic diseases. *Journal of Community Health*, *44*, 400–411.

Steinert, T., Lepping, P., Bernhardsgrutter, R., *et al.* (2010). Incidence of seclusion and restraint in psychiatric hospitals: A literature review and survey of international trends. *Social Psychiatry and Psychiatric Epidemiology*, *45*, 889–987.

Tasmania, D. H. S. (2017). CCP Standing Order 10 – Chemical Restraint. https://www.dhhs.tas.gov.au/__data/assets/pdf_file/0008/252755/CCP_Standing_Order_10_-_Chemical_Restraint.pdf. Accessed: 18 March 2019.

Taylor, D., Yap, C. Y. L., Taylor, S. E., *et al.* (2016). Midazolam-droperidol, droperidol, or olanzapine for acute agitation: A ranomized clinical trial. *Annals of Emergency Medicine*, *69*, 318–326.

The Joanna Briggs Institute (2014). *Joanna Briggs Institute Reviewers' Manual*, 2014th edn. Adelaide, South Australia, Australia: The University of Adelaide.

Tingleff, E. B., Bradley, S. K., Gildberg, F. A., Munksgaard, G. & Hounsgaard, L. (2017). "Treat me with respect". A systematic review and thematic analysis of psychiatric patients' reported perceptions of the situations associated with the process of coercion. *Journal Psychiatry and Mental Health*, *24*, 681–698.

TREC Collaborative Group (2003). Rapid tranquillisation for agitated patients in emergency psychiatric rooms: A randomised trial of midazolam versus haloperidol plus promethazine. *British Medical Journal*, *327*, 708.

Walther, S., Moggi, F., Horn, H., *et al.* (2014). Rapid tranquilization of severely agitated patients with schizophrenia spectrum disorder: A naturalistic, rater-blinded, randomized, controlled study with oral haloperidol, risperidone, and olanzapine. *Journal of Clinical Psychopharmacology*, *34*, 124–128.

Weiland, T. J., Ivory, S. & Hutton, J. (2017). Managing acute behavioural disturbances in the emergency department using the environment, policies and practice: A systematic review. *Western Journal of Emergency Medicine*, *18* (4), 647–661.

Yap, C. Y. L., Taylor, D., Knott, J. C., *et al.* (2017). Intravenous midazolam–droperidol combination, droperidol or olanzapine monotherapy for methamphetamine-related acute agitation: Subgroup analysis of a randomized controlled trial. *Addiction*, *112*, 1262–1269.

© 2019 Australian College of Mental Health Nurses Inc.